Code No: 152AG **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, September/October - 2021
ENGINEERING GRAPHICS
(Computer Science and Engineering)

Time: 3 Hours **Max Marks: 75**

Answer any three questions
All questions carry equal marks
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1.a) Draw a hyperbola having two asymptotes perpendicular to each other and passing through a point P at a distance of 30 mm from one asymptote and 36 mm from other. Draw a normal and tangent at any convenient point.

b) Construct a diagonal scale of R.F = 1: 4000 to show meters and long enough to measure 500 meters. Indicate a length of 379 meters. [13+12]

2.a) Draw the projections of the following points on a common reference line keeping the distance between their projectors 30 mm apart.
i) Point A is lying 70 mm above the H.P. and on the V.P.
ii) Point B is lying on the H.P. and 50 mm behind the V.P
iii) Point C is lying 70 mm above the H.P. and on the V.P
iv) Point D is lying on the H.P. and 50 mm in front of the V.P.
v) Point E is lying 70 mm below the H.P. and 50 mm behind the V.P

b) A 70 mm long line PQ, has its end P 20 mm above the H.P. and 30 mm in front of the V.P. The line is inclined at 45° to the H.P. and 30^0 to the V.P. Draw its projections. [10+15]

3. A square pyramid of base side 30 mm and altitude 50 mm lies on one of its triangular faces on the H.P with its axis parallel to the VP. It is cut by a vertical plane inclined at 30^0 to the VP and meeting the axis at 40 mm from the vertex measured in the plan. Draw the top view, sectional front view and the true shape of the section. [25]

4. Vertical cylinder of 70 mm diameter is penetrated by another cylinder of 45 mm diameter the axis of which is parallel to both HP and VP. The two axes are 9 mm apart. Draw the projection showing curves of intersection. [25]

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5. Draw the isometric view of the machine parts shown in figure 1. (All dimensions are in mm) [25]

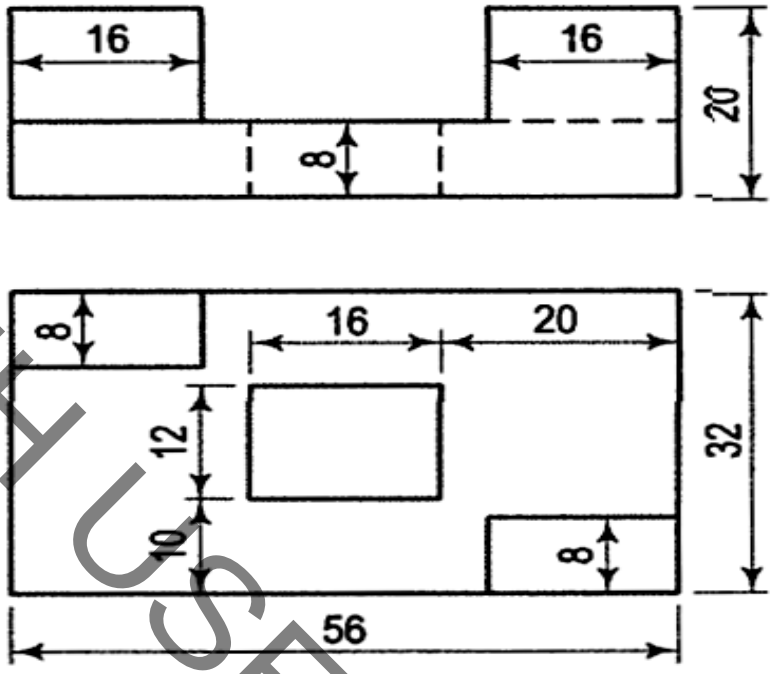


Figure: 1

6. Draw the orthographic projections of the machine part shown in figure 2: a) Front view b) Top view c) Side view. All dimensions are in mm. [25]

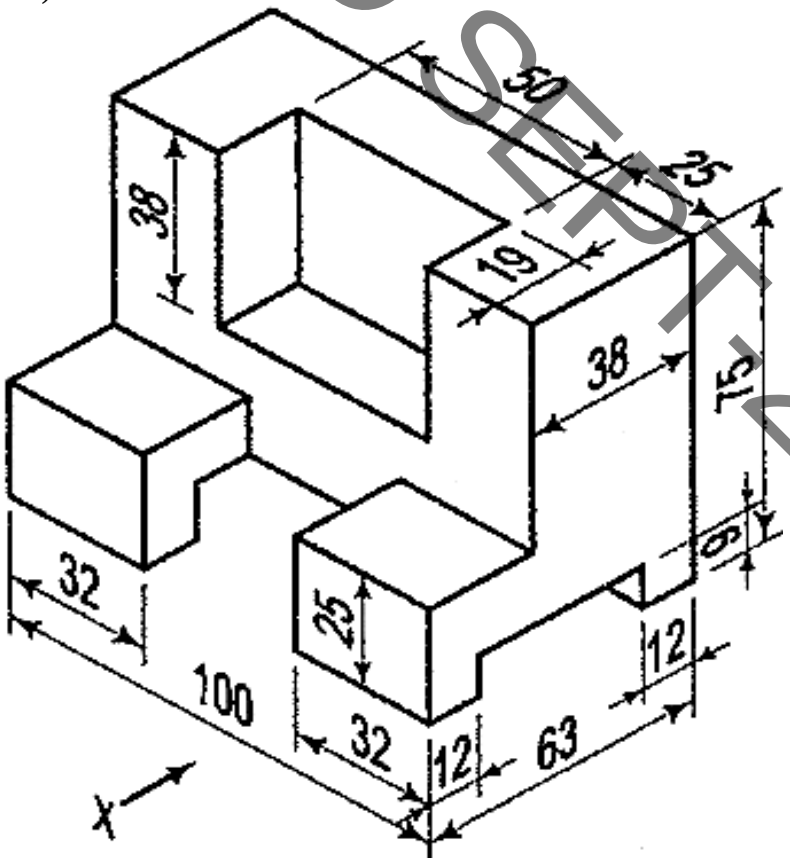


Figure: 2

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